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Modeling Real-Time Bicep Muscle Fatigue With IMU Sensors Using LSTM Regression

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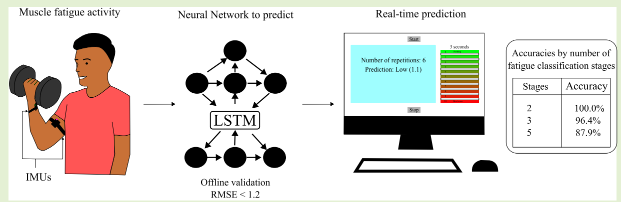
***Abstract*—Muscle fatigue is a leading contributor to musculoskeletal disorders (MSDs), which cause significant workplace disability and absenteeism. The biceps brachii are especially susceptible in manual labor due to prolonged use during tasks like lifting and carrying. Early detection of fatigue can help prevent MSDs by enabling timely interventions, such as workload adjustments or the use of assistive devices. This study presents a real-time, noninvasive, and cost-effective system for modeling biceps muscle fatigue during dumbbell curl exercises, which simulate industrial lifting movements. Two inertial measurement units (IMUs) captured arm kinematics, and a long short-term memory (LSTM) neural network was trained to predict fatigue levels on a 0–7 scale using participants' perceived exertion ratings. After offline training and optimization, the model was implemented in real-time and evaluated across 18 experiments, achieving a root-mean-squared error (RMSE) of 1.2. Accuracy was assessed using RMSE thresholds (±5, ±3.3, and ±2), reaching 100% at ±5 and exceeding 87% for offline and online implementations. Compared to previous studies, this approach achieved higher accuracy with fewer sensors and demonstrated reliable real-time performance, highlighting its potential for workplace fatigue monitoring and injury prevention.**



***Index Terms*—Inertial measurement unit (IMU), long short-term memory (LSTM) neural network, muscle fatigue, real-time monitoring.**

I. Introduction

MUSCLE fatigue is the gradual decline in a muscle's ability to generate force, reducing an individual's level of physical strength. This condition generally results from repeated and sustained physical activity [1], [2]. Fatigue serves as a defense mechanism in humans, preventing metabolic crises and protecting the integrity of muscle fibers. The accumulation of muscle fatigue over time has severe consequences, including musculoskeletal disorders (MSDs), chronic fatigue syndrome, decreased immune system function, and long-term reductions in strength and motor control [3].

MSDs are one of the main concerns in occupational health. Common examples of MSDs include back pain, carpal tunnel syndrome, tendinitis, and arthritis. They can lead to discomfort, pain, and decreased mobility, and significantly cause disability and absenteeism in the workplace [4]. These disorders contribute both directly and indirectly to the economic burden. In United States, over 80% of workers are affected by some form of MSD [5]. Similarly, in United Kingdom (U.K.), 470 000 workers reported having MSDs between 2020 and 2021 [6]. In Brazil, according to the Global Health Data Exchange [7], the prevalence of people with MSDs was 56 512.688 in 2021.

The biceps brachii are particularly vulnerable to MSDs during manual handling tasks, such as repetitive lifting and carrying boxes, equipment, or other heavy materials. These tasks are common across various industries, including construction, healthcare, warehousing, and manufacturing. In the

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U.K., 45% of MSD cases involved muscle fatigue in the upper limbs and neck [6]. Biceps tendonitis is a common MSD characterized by tendon inflammation between the biceps and the shoulder. This is often a result of repetitive or overuse from manual lifting activities [8].

Detecting muscle fatigue is critical for preventing long-term health consequences among blue-collar workers. Fatigue assessment methods can be categorized as subjective or objective. Objective methods monitor physiological parameters such as heart rate and electromyography (EMG) [3], while subjective methods rely on self-reported measures, including questionnaires and perception indexes [9]. A widely used subjective measure is the Borg rate of perceived exertion (RPE) scale [10], in which individuals rate their perceived level of physical exertion on a numerical scale during activity. This scale has been validated as a reliable indicator of muscle fatigue across various physical tasks, including manual handling activities performed by workers in real-world settings [11], [12].

Recent studies have explored the inclusion of machine learning (ML) for muscle fatigue detection. These studies have trained supervised ML models to estimate discrete stages of muscle fatigue from sensor measurements during physical activity. In [13], resistive sensors embedded in a soft glove were used to detect two levels of hand fatigue. A real-time random forest (RF) classifier was implemented, achieving an accuracy of 95.7%. In [14], a fabric-based resistive strain sensor was used to monitor the thickness variation of the biceps, and EMG was used to label the data into four stages of fatigue. A support vector machine (SVM) classifier was trained with up to 83.3% accuracy. In [15], the EMG signal in the biceps was decomposed using a discrete wavelet transform, and temporal and spatial features were extracted. Using an SVM model, high feature separability was found along what was hypothesized to be nonfatigue and fatigue classes, with up to 99.5%. In [16], a heart rate sensor, sEMG, and IMU were used to estimate four stages of lower body muscle fatigue. The fatigue labels were derived from blood lactate levels and Borg RPE ratings. This study achieved up to 96.5% accuracy using logistic regression (LR).

In [17], an optical fiber sensor (OFS) for angle measurement, EMGs, and IMUs were used to estimate three fatigue states based on Borg RPE ratings for the biceps. An accuracy of 96.8% was achieved with a light gradient-boosting machine (LGBM) classification model. In [18], a heart rate sensor and an IMU were used to detect binary fatigue classifications of the biceps. The fatigue classification was labeled by Borg RPE ratings and heart rate measurements. Up to 94% classification accuracy was achieved with a feedforward neural network (FNN). In [19], EMG and IMU data collected during scaffold-building activities, involving various upper body muscles, were used to predict aerobic fatigue threshold (AFT), which was used to determine five stages of fatigue labels, achieving up to 92.3% accuracy using a decision tree (DT). Table I shows the literature results. This aims to identify and classify fatigue during physical activity. However, they primarily use time-independent models, which may impact real-time performance.

TABLE I
LITERATURE REVIEW OF FATIGUE PREDICTION STUDIES

Ref.	Input Sensor	Fatigue Label	Target Muscle	Fatigue Stages	Best Model	Best Accuracy(%)
[15]	EMG	EMG	biceps	2	SVM	99.5
[18]	heart rate sensor, IMU	heart rate, Borg RPE	biceps	2	FNN	88.0
[14]	fabric-based resistive strain sensor	EMG	biceps	3	SVM	83.3
[17]	optical fiber sensor, EMG, IMU	EMG Borg RPE	biceps	3	LGBM	96.8
[16]	heart rate sensor, EMG, IMU	blood lactate, Borg RPE	lower body	4	LR	96.5
[19]	EMG, IMU	AFT	upper body	5	DT	92.3
Ref.	Input Sensor	Fatigue Label	Target Muscle	Fatigue Stages	Best Model	RMSE
[20]	EMG 3D depth sensor	MDF	Biceps Deltoid	2	LSTM	0.09 0.17
[21]	IMU	RPE	lower body	Continous (1-10)	Hybrid	0.21

The classification results of these studies are promising, but two main limitations were identified. First, most of these models have not been evaluated for their real-time performance. Most models operate in an offline setting, i.e., given a segment of sensor data, the model is applied retroactively to predict fatigue. The nature of the models (SVM, LR, LGBM, FNN, and DT) is all time-independent, which means the model is applied to each segment of the input data with no regard to sequential or temporal order. This limits the real-time effectiveness of these models [22]. To build systems that can be deployed in occupational settings, it is necessary to expand the models' capacity to understand the temporal nature of the signals and explicitly evaluate their real-time performance.

The second limitation is the oversimplification of fatigue classification categories. Muscle fatigue is a gradual process, but existing models are trained to classify fatigue into discrete stages. From an ML perspective, this classification setup is not representative of the nature of the task. For example, in a simple three-stage classification (low, moderate, and high fatigue), the model is penalized equally for classifying moderate fatigue as low as it is for classifying high fatigue as low. In reality, a setup that emphasizes learning the overall trend of fatigue is desired as opposed to learning strict boundary choices between discrete stages. Posing this task as a classification problem is an ineffective way to model muscle fatigue, as muscle fatigue is inherently a regression problem [23].

Another approach involves the use of deep learning (DL) algorithms capable of modeling temporal dependencies in physiological signals. In [20], a long short-term memory (LSTM) network trained with synchronized EMG and motion data was employed to predict muscle fatigue. The EMG median frequency (MDF) was used to label the data, and the model achieved a root-mean-squared error (RMSE) of 0.09 and a mean absolute error (MAE) of 0.07 for the biceps brachii, as well as an RMSE of 0.17 and a MAE of 0.07 for the lateral deltoid. Although EMG signals are most commonly used in these applications, they present limitations for long-term monitoring due to factors such as skin sweating and electrode displacement, which can degrade signal quality [24]. Similarly, in [21], five IMUs positioned on the lower limbs were used to predict exercise-induced fatigue through a forecasting network. This network combined a Transformer with a critic composed of three convolutional layers and one linear layer, and a DeepConvLSTM, achieving an RMSE of 0.21 and an accuracy of 83% (±0.5) in real time.

Considering the limited number of studies that estimate upper-limb fatigue in real time, and the importance of this capability for preventing long-term consequences in workers, this study aimed to develop a real-time system for modeling biceps muscle fatigue during dumbbell curl exercises using a single type of sensor. The proposed system employs two IMUs as input, ensuring a noninvasive and cost-effective approach. While most existing real-time implementations focus on lower limb fatigue estimation, the proposed approach extends this capability to the upper limb. To address the limitations of previous works, a time-dependent ML model was trained for regression, enabling a more accurate representation of the gradual progression of fatigue in real time. This real-time implementation is particularly relevant, as it allows immediate feedback and potential intervention during physical activity, critical for preventing injuries, optimizing training, and supporting rehabilitation or occupational tasks.

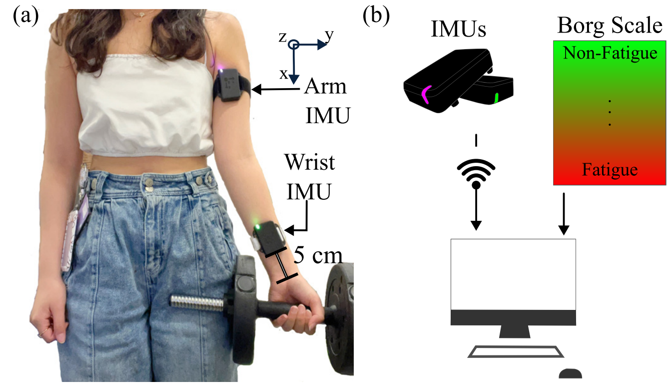

Fig. 1. (a) Wrist and arm IMUs placement. (b) System inputs: The IMUs wirelessly transmit motion data to a computer, while subjective fatigue levels are recorded using the Borg scale.

II. Methods

The development of the model followed three main stages: data collection, data preprocessing, and model training. First, the data collection phase established the experimental protocol, described the equipment used, and detailed participant recruitment. Next, in the data preprocessing stage, the IMU signals were preprocessed, time series segmentation was performed, the regression target variable was defined, and relevant features were extracted. Finally, the model training phase involved designing the model architecture, implementing the training strategy, and selecting the most pertinent features for optimal performance.

A. Data Collection

1) Experimental Protocol: For the experiment, two IMUs are securely strapped to the participant's arm: one on the forearm, approximately 5 cm from the wrist crease, and another on the upper arm near the biceps. The placement of both IMUs is illustrated in Fig. 1(a). Each IMU was mounted with its local coordinate system aligned consistently across participants. In particular, the sensor's x-axis was oriented laterally (pointing outward from the arm), the y-axis anteriorly (forward, in the sagittal plane), and the z-axis superiorly (upward along the arm's longitudinal axis), as illustrated in Fig. 1(a). This consistent body-to-sensor alignment ensured repeatability across participants and reproducibility of the measurements.

IMUs were chosen as the input signal due to their robustness, cost-effectiveness, and lightweight nature. Previous studies [16], [17], [18], [19] have explored the use of IMUs for fatigue-related applications, often in combination with other modalities such as EMG or OFS. For instance, some works reported high accuracy when IMU signals were combined with other signals, while others highlighted the potential of IMUs as standalone indicators of fatigue. Considering that, this study focuses exclusively on off-the-shelf IMUs to investigate their capability as a noninvasive and practical tool for muscle fatigue estimation. Physiological measurements, such as heart rate and EMG, were avoided due to their high variability. For example, heart rate can be influenced by various mental and physical factors (e.g., emotions and ambient temperature) beyond muscle fatigue. Cost-effective OFS prototypes face limitations in terms of repeatability and susceptibility to misalignments, which affect the sensor's sensitivity and restrict their applicability in real-world scenarios.

Each participant was instructed to perform repeated dumbbell curls at a comfortable and consistent pace, with minimal rest between repetitions. The dumbbell curl exercise was selected because it directly activates the biceps muscle and represents a simple yet effective lifting movement [17]. Prior to the experiment, participants were familiarized with the Borg CR10 scale, which includes half-intervals. The Borg CR10, a variation of the Borg RPE scale, has been validated as a reliable indicator of muscle fatigue during manual upper limb tasks [11]. This scale was used to obtain the ground truth (regression target variable), as it provides a straightforward and intuitive estimation of perceived muscle fatigue. During the experiment, participants reported their perceived exertion every 5 s using the scale. The session concluded when the participant reached level 7 ("very high") on the Borg scale. Both IMU data and Borg scale ratings were collected and processed for subsequent analysis and prediction generation, as illustrated in Fig. 1(b).

2) Equipment: The dumbbell used in the experiment is from Anchor's Adjustable Dumbbell Weights. It consists of a lightweight bar with two 1.5-kg plates on the end, totaling approximately 3 kg. The IMUs used in the experiment are the x-IMU3 wireless (x-io Technologies, U.K.). These IMUs record 3-D accelerometer and gyroscope measurements, as well as Euler angles (roll, pitch, and yaw). The IMUs positioned on the wrist and arm were time-synchronized and calibrated to ensure consistent and accurate measurements across both sensors.

3) Participant Selection: Participants are recruited on a volunteer basis. To be included in the data collection, participants

TABLE II
TRAINING DATA DEMOGRAPHIC DISTRIBUTION

Gender	Participants	Age	BMI	Experiments
Male	41	29.5 ± 9.3	24.0 ± 2.6	74
Female	35	26.0 ± 5.4	24.1 ± 4.6	66
Total/Average	76	27.8 ± 7.8	24.1 ± 3.7	140

TABLE III
TIMESERIES MEASUREMENTS COLLECTED FOR WRIST AND ARM IMUS, INCLUDING DIFFERENCES IN EULER ANGLES BETWEEN THE IMUS

IMU		Measurement
Arm, Wrist	Gyroscope	x, y, z, magnitude (°/s)
	Accelerometer	x, y, z, magnitude (g)
	Euler Angles	roll, pitch, yaw (°)
	Euler Angles Differences	roll, pitch, yaw (°)

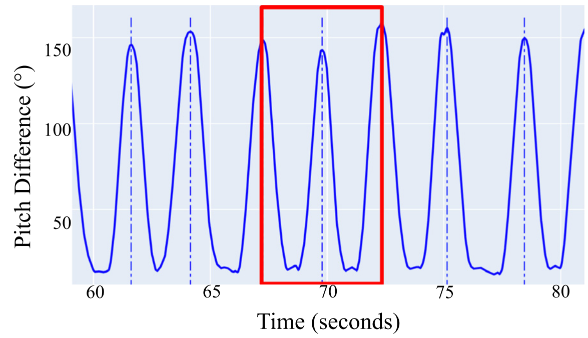

Fig. 2. Pitch difference between arm and wrist IMUs. Dotted vertical lines delineate points of maximum arm flexion (peaks). Data are segmented into overlapping windows between three consecutive peaks. An example of such a segment is boxed in red.

are required to be in good physical health and between 18 and 40 years of age. Participants are excluded if they have a history of musculoskeletal, cardiovascular, neurological, or other injuries/health conditions affecting the muscles, bones, or joints of the upper body. Signed consent forms were obtained from the participants after they were informed of the experimental protocol. The ethical committee of the University of the West of England approved this protocol.

In this study, one experiment corresponded to a single recording session for one arm. On average, each participant completed two sessions (one per arm), except for 15 participants who completed only one. A total of 76 participants were recruited, and data were collected from 140 experiments. Each experiment consisted of approximately 38 dumbbell curl repetitions, resulting in a total of 5149 recorded repetitions. The demographic information of the participants is presented in Table II.

B. Data Preprocessing

1) IMU Signals: The IMU data is streamed at 50 Hz via a user datagram protocol (UDP) connection to the system. For each IMU, gyroscope, and accelerometer, measurements are collected in the x, y, and z axes, as well as an additional magnitude dimension $(x^2 + y^2 + z^2)^{1/2}$ for a normalized accelerometer signal [25]. However, the three Euler angles (roll, pitch, and yaw) are collected for each IMU, as well as the difference between each corresponding Euler angle from the arm IMU and the wrist IMU, e.g., $\text{pitch}_{\text{difference}} = \text{pitch}_{\text{arm IMU}} - \text{pitch}_{\text{wrist IMU}}$. The system computes the magnitudes and Euler angle differences in real-time from each timestep of the IMU sensor measurements. In total, 25 time series signals are obtained for each experiment, as shown in Table III.

For each of these signals, a rolling average filter with a 100-ms sliding window was applied to reduce noise. This choice of filter and window size was made after visually inspecting the signals generated during an initial trial run of the experiment. During this inspection, it was observed that the 100-ms window effectively smooths out short-term fluctuations caused by noise while preserving the essential patterns (e.g., peaks, troughs, and overall trends) of the signals.

2) Time Series Segmentation: Next, these time series measurements were segmented into smaller time windows to analyze the patterns of motion as fatigue progresses. To accomplish this, the time stamps for each biceps curl repetition are first identified. Each bicep curl is defined as the movement from maximum arm extension (when the arm is relaxed) to maximum arm flexion (when the elbows are bent and the dumbbells are raised toward the shoulder) and back to maximum arm extension. This motion is best captured through the Euler angle pitch difference between the arm and wrist IMUs, as shown in Fig. 2. Troughs indicate points of maximum arm extension; peaks indicate maximum arm flexion. Previous works in bicep curl IMU data have used peak-to-peak [17] or trough-to-trough [26] segmentation. Overlapping windows between 3 consecutive peaks are used, as boxed in Fig. 2. This method captures more contextual information in the pattern of motion and is more robust to data anomalies, such as weight shifts in between bicep curl repetitions. This is because each time window considers the motion across three bicep curl repetitions.

3) Regression Target Variable: After segmenting the data into time windows, the regression targets were derived for each window. This process involves assigning a Borg value to every timestamp in the data. Since participants provide a Borg value every 5 s during the experiment, linear interpolation [17] is used to estimate the Borg values for the timestamps between their responses. For each time window, the mean of these interpolated Borg values is calculated and used as the regression target.

4) Feature Extraction: To prepare data for model training, relevant features are extracted from each time window. After reviewing existing literature [25], [27] on feature extraction from IMU data, 11 key features are identified, as detailed in Table IV. For each of the 25 time series signals from the IMU data, these 11 features are extracted, for a total of 275 features.

To ensure consistency across different experiments, the features are normalized within each experiment using the features from the first time window as [17]. In the real-time system, once the participant completes the first three repetitions, the features extracted from that initial time window are used as the baseline for normalizing the features in all subsequent time windows within the same experiment.

TABLE IV
FEATURES COMPUTED FOR EACH SIGNAL

Feature	Computation
Mean (Average)	$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$
Standard Deviation	$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2}$
Skewness	$\text{Skewness} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i-\mu}{\sigma}\right)^3$
Kurtosis	$\text{Kurtosis} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i-\mu}{\sigma}\right)^4 - 3$
Range	$\text{Range} = \max(x) - \min(x)$
Maximum	$\text{Maximum} = \max(x)$
Minimum	$\text{Minimum} = \min(x)$
Root Mean Square (RMS)	$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
Lag 1 Autocorrelation	$\text{Autocorrelation} = \frac{\sum_{i=1}^{N-1}(x_i-\mu)(x_{i+1}-\mu)}{\sum_{i=1}^{N}(x_i-\mu)^2}$
Total Power	$\text{Total Power} = \sum_{i=1}^{M} P(f_i)$
Dominant Frequency	$\text{Dominant Frequency} = f[\text{peaks}[0]]$

The sequence of features extracted from all time windows in an experiment is concatenated to get the final input vector with dimensions num_time_windows $\times$ num_features. In order to train the model to perform regression, the corresponding target vector is prepared with the sequence of Borg averages (see Section II-B.3). A padding of -1 is used for both the input and target vectors using the maximum sequence length across all the training data to ensure equal dimensions of all data points during training.

C. Model Training

1) Model Architecture: In this study, a recurrent neural network (RNN) [28] model is used to capture the temporal nature of the data. The input layer takes in the sequence of features from a single experiment. This input is passed through a single LSTM [29] layer consisting of 64 units. Finally, it is passed through a fully connected layer with a single output unit to produce the final prediction for each time step.

Previous works [30], [31] have shown the advantages of LSTMs applied to feature-based time series data, rendering both higher prediction accuracy and lower training time compared to other ML models. Proof-of-concept studies have been conducted on IMU data with LSTMs, such as in gait phase estimation [32] and human activity classification [33].

2) Training Strategy: To train the model, the log-cosh loss function [34] was used, as it provides smoother regression predictions and greater robustness to outliers compared to other loss functions. An Adam optimizer [35] was employed with a two-stage learning-rate scheduling: 1) training for 200 epochs with a learning rate of 0.001, and 2) retraining for another 200 epochs with a reduced learning rate of 0.0001, using the weights from the best model in the first stage. In both stages, the model performance was monitored based on the lowest RMSE validation loss, and the corresponding weights were used for the next stage/final model. The number of timesteps to the LSTM input was defined as the maximum sequence length among all training samples; all inputs were padded or truncated at the end to match this length. The model used a linear activation at the output layer and was trained with a batch size of 128. No dropout or regularization techniques were applied. These parameters were determined experimentally [36].

3) Feature Selection: To reduce the number of features in each timestep of the input sequence, a greedy forward feature selection [37] is performed, which has been shown to improve prediction accuracy and training time.

The process starts by evaluating each feature individually to determine its predictive power. The RMSE was used as the evaluation metric. The feature that results in the lowest average RMSE across all validation datapoints is selected as the first feature.

Next, features are iteratively added to the selected set. In each interaction, the feature that, combined with those already selected, produces the lowest average RMSE was added. This process continues, adding one feature at a time, until no additional features reduce the average RMSE.

4) Offline Cross-Validation: The 76 participants in the training pool were divided into eight roughly equal groups for eightfold cross-validation. Four folds included nine participants for validation and 67 for training, while the remaining four folds included ten participants for validation and 66 for training [38], [39].

Using the training strategy described above, a model is trained on each training set. The RMSE and accuracy are then computed on the corresponding validation set. To calculate the prediction accuracy and compare it with the literature, three RMSE values are considered based on the number of classification stages in previous studies (±5, ±3.3, and ±2). The ten levels of the Borg scale are divided into stages, classified by each algorithm.

An overview of the methodology is outlined in Fig. 3. However, the integrated system for real-time application is available at https://github.com/erinliu1/Muscle_Fatigue

III. RESULTS

A. Offline Cross-Validation

Although the main objective is real-time implementation, a preliminary offline analysis was conducted to train, validate, and optimize the model under controlled conditions. This step allowed for hyperparameter tuning and feature selection without the latency introduced by live data transmission, thereby ensuring that the online version could achieve optimal performance once deployed.

Using greedy forward feature selection, a subset of seven features out of the original 275 features is selected. They are listed in Table V. This selected subset of features resulted in an average RMSE of 1.13 $\pm$ 0.60. This indicates that the model's predictions were, on average, within 1.13 $\pm$ 0.60 of the actual Borg values provided by the participants.

B. Real-Time Testing

1) Final Model: To train the final model for real-time testing, the training data is split by randomly selecting data from 10 individuals to serve as validation data. The remaining data

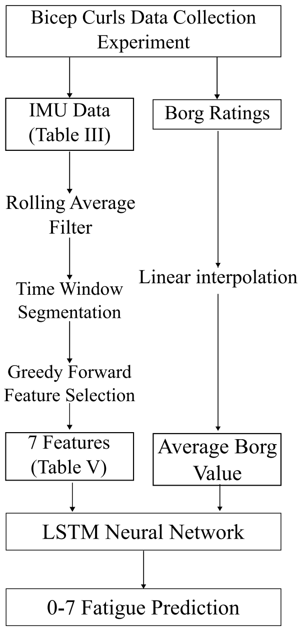

Fig. 3. Flowchart of the muscle fatigue prediction framework. The process begins with data collection from an IMU and Borg ratings during a bicep curls experiment. IMU data undergoes preprocessing, feature selection, and extraction of seven key features, while Borg ratings are interpolated. The features and average Borg value serve as inputs to an LSTM neural network, which predicts fatigue levels on a scale of 0–7.

TABLE V
SELECTED FEATURES FROM THE GREEDY FORWARD FEATURE SELECTION PROCESS

Signal	Selected Features
Wrist Gyroscope Y	Total Power
Wrist Gyroscope Z	Kurtosis
Arm Gyroscope X	Maximum
Arm Gyroscope Y	Lag 1 Auto-correlation
Arm Gyroscope Z	Minimum
Arm Accelerometer Magnitude	Mean
Difference Pitch	Lag 1 Auto-correlation

TABLE VI
TEST DATA DEMOGRAPHIC DISTRIBUTION

Gender	Participants	Age	BMI	Experiments
Male	5	27.0 ± 5.3	21.6 ± 3.6	8
Female	5	30.2 ± 6.1	22.6 ± 1.9	10
Total/Average	10	28.6 ± 5.6	22.1 ± 2.8	18

TABLE VII
REAL-TIME RMSE AND ACURRACIES RESULTS PER PARTICIPANT

			Accuracy (%)		
Individual	Experiment number	RMSE	± 2	± 3.3	± 5
1	1	0,99	100,00	100,00	100,00
	2	1,76	69,51	93,90	100,00
2	1	1,32	87,50	100,00	100,00
3	1	1,81	68,42	100,00	100,00
	2	1,77	74,07	100,00	100,00
4	1	0,36	100,00	100,00	100,00
	2	0,83	100,00	100,00	100,00
5	1	1,61	96,15	100,00	100,00
	2	1,17	88,46	100,00	100,00
6	1	0,46	100,00	100,00	100,00
7	1	0,96	100,00	100,00	100,00
	2	2,54	51,85	62,96	100,00
8	1	0,87	100,00	100,00	100,00
	2	0,82	94,29	100,00	100,00
9	1	1,15	87,10	100,00	100,00
	2	0,78	100,00	100,00	100,00
10	1	0,80	100,00	100,00	100,00
	2	1,66	73,91	100,00	100,00

from 66 individuals in the training pool is used to train the model. It followed the same model architecture and training strategy detailed in Sections II-C.1 and II-C.2. The model is trained using the seven features selected from the greedy forward feature selection process. The model checkpoint corresponding to the epoch with the lowest validation loss in Keras format is saved, as well as the scaler used for feature normalization in PKL format. These saved files are then used for real-time inference.

2) Test Participants: To test the real-time performance of the model while avoiding any training bias, ten participants distinct from the 76 individuals used in model training/validation are recruited. A summary of the demographic distribution of the participants is shown in Table VI. These participants performed the same experiment as before (see Section II-A.1). Two participants completed the test using only one arm. Participants performed an average of 39 repetitions (range $Y–Z$) during real-time experiments, totaling 696 repetitions across all trials.

3) Real-Time Inference: To employ the real-time model, an integrated system is built for model inference and new data collection.[1]

Once the x-IMU sensors are connected to the same Wi-Fi network as the computer running the system, it is possible to start recording data. When the system detects a bicep curl repetition (see Section II-B.2), it extracts the features from the IMU data and feeds them to the trained model, which then outputs its numerical Borg value prediction. Concurrently, the experimenter can ask participants to provide their Borg ratings every 5 s, and the system will allow the experimenter to save these responses in real-time with the IMU data. These ratings are used to verify the accuracy of the model's predictions in the real-time experiments. This data collection integration also allows for easy model retraining/fine-tuning with new data.

Table VII reports RMSE and accuracy for each participant in the real-time implementation. Across the 18 real-time experiments, the mean RMSE was 1.20 ± 0.56, indicating that on average the model predictions deviated from the actual Borg values of the participants by 1.20 ± 0.56. Fig. 4 shows the distribution of RMSEs from both offline and real-time experiments.

Also, the average inference time (the average time between the system's detection of a bicep repetition and its prediction output) was computed to be 38 ms on a personal computer (PC). Also, this value was calculated on a small single-board computer (SBC) Raspberry Pi 4 (Raspberry Pi, U.K.), with

[1] In the real-time system, it displayed a description for fatigue along with the numerical prediction. For predictions less than 3, the description is "low"; for predictions between 5 and 7, the description is "moderate"; for predictions greater than 7, the description is "high."

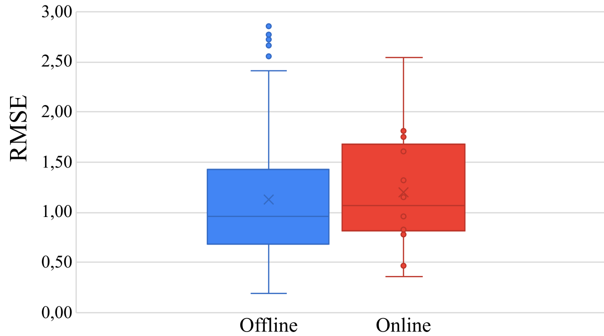

Fig. 4. Box and whiskers plot of the RMSE for the offline and real-time experiments.

TABLE VIII
COMPARISON OF THE ACCURACY (ACC) RESULTS REPORTED IN THE LITERATURE WITH THOSE OBTAINED IN THE PRESENT WORK FOR ONLINE AND OFFLINE PREDICTIONS

		Literature			Actual work	
Stages	RMSE	Input Sensors	Algorithm	Acc (%)	Acc (%) Offline	Acc (%) Online
2	±5	EMG [15]	SVM	99.5	100.00	100.00
		Heart rate, IMU [18]	FNN	88.0		
3	±3.3	Strain [14]	SVM	83.3	98.89	96.41
		Angle, EMG, IMU [17]	LGBM	96.8		
5	±2	EMG, IMU [19]	DT	92.3	88.15	87.93

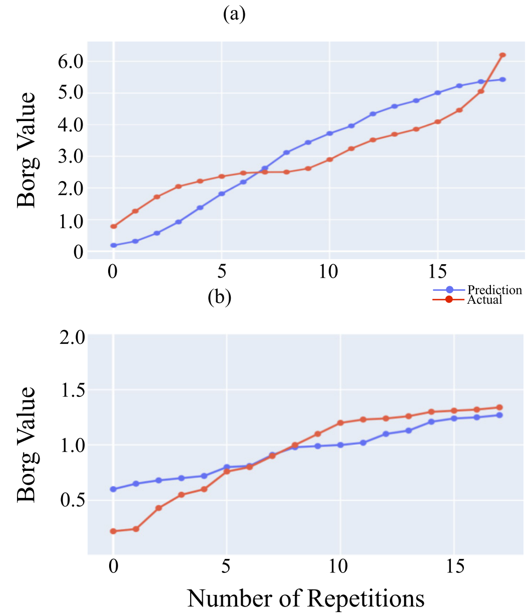

Fig. 5. Graph shows model's predictions versus participant's Borg responses for two real-time experiments. (a) Participant performed 19 bicep curl repetitions with the standard 3-kg dumbbell. The RMSE is 0.78. (b) Participant performed 18 repetitions with a lowered 1.4-kg dumbbell. The RMSE is 0.75.

a mean of 268 ms. Both results are much smaller than the minimum time it takes for someone to perform a bicep curl, demonstrating the real-time feasibility of the system.

Finally, Table VIII presents accuracy results from the literature alongside those obtained in this study (both offline and online) for each RMSE. For an RMSE of ±5, accuracy reached 100%. For the other two RMSE values evaluated, accuracy remained above 88% for the offline and online implementations.

IV. DISCUSSION

The model's predictions largely have the same fatigue trend as the actual Borg ratings. For instance, in Fig. 5(a), an example of the results was illustrated from a test experiment. From the graph, it is possible to see that although the actual predictions are not precisely the same as the Borg ratings, the overall increase in fatigue is captured accurately. Thus, the initial hypothesis was validated: the regression approach reduces egregious errors compared to classification. In classification-based models with low resolution, predictions may be limited to only a few stages, leading to extreme misclassification, such as predicting high fatigue when it is actually low and vice versa [15], [18].

To control for the confound that the model is just learning to increase fatigue linearly with time, participants were tested with a lighter 1.4-kg dumbbell. The expectation is that the participants' fatigue trends should remain consistently low due to the lightness of the dumbbell. In Fig. 5(b), it is shown that the model accurately predicted a stagnant fatigue trend, demonstrating that it is not just linearly increasing fatigue over time.

From these results, it was hypothesized that the model is learning to correlate compensatory motions in the bicep curl motion with increasing fatigue [40]. If the participant is demonstrating consistent patterns of motion, such as in Fig. 5(b), the model predicts consistent low fatigue. However, when the participants enter fatigue, their motion patterns alter to compensate for their bicep fatigue, such as larger shoulder movements, wider patterns of motion, etc. Thus, the model detects these compensatory motions as an indication of increasing fatigue, as in Fig. 5(a).

To better understand the nature of the learned model and the data, the outlier experiments were examined, where the model consistently under-predicted or over-predicted the participants' fatigue. It was hypothesized that the subjective nature of the Borg scale may contribute to this discrepancy. Different people have different thresholds for what they perceive as fatigue. For instance, an individual who often goes to the gym and lifts weights will likely have a higher tolerance for fatigue than an individual who does not often experience lifting heavy objects [41]. Thus, the presence of compensatory motion may not directly correlate with the perception of fatigue [42]. Ultimately, both of these are just proxies for the actual presence of muscle fatigue, which could only be detected with physiological measures.

Although the study included fewer female than male participants, evidence in the literature suggests that sex-related differences in fatigability are not significant. For example, Hubal et al. [43] reported no significant differences in fatigability between men and women during maximal eccentric exercise of the elbow flexors. Similarly, studies on dynamic exercise have not found consistent sex-related differences in fatigue [44]. Therefore, this variable was not considered during the training process.

A different approach is to use more objective methods of labeling fatigue, such as EMG or AFT [23], [19]. Furthermore, even when subjective measures of fatigue are used, obtaining data from a larger and more diverse population of participants should compensate for some of the subjective differences in perceived exertion. Another possible remedy could be to normalize model predictions based on each individual's baseline or maximum level of physical strength [42].

Another consideration is the length of the experiments. The number of data points that an LSTM can take in a sequence is primarily determined by the available memory (RAM and/or GPU). The maximum experiment length for this study is currently capped at 300 bicep curl repetitions because it was the most repetitions any participant did continuously. In a real-world occupational setting, it is anticipated that the user will take breaks from the manual handling task, so for memory efficiency, it is possible to integrate into the system the ability to detect when the user is taking a break and discard portions of its memory accordingly.

In recent years, interest in estimating muscle fatigue has increased. However, to the best of our knowledge, no real-time implementation has been performed, making this the main contribution of this article. The accuracy results indicate that the proposed approach outperforms most previous studies, except for that of Bangaru et al. [19], who used a DT to classify five fatigue stages, achieving an accuracy of 92%. In comparison, this study obtained an accuracy of 88% for offline implementation.

Considering that the presented system utilizes only two IMUs, it can be regarded as a more simplified approach than those found in the literature. Although most other studies reported high accuracy (above 92%), the algorithms used, such as SVM [14], [15], LR [16], LGBM [17], and FNN [18], do not consider the temporal order of the data, i.e., the sequential nature of the signals is ignored. In contrast, the proposed system employs an LSTM neural network, which better captures fatigue patterns as they develop over time, making real-time implementation more effective.

Finally, when compared with the work of Jiang et al. [21], which also employed an LSTM-based algorithm, the proposed model achieved slightly higher RMSE values (1.13 for offline and 1.20 for online prediction) than theirs (0.09 for the biceps and 0.17 for the deltoid). However, their approach relied on EMG-derived features for the estimation. Considering the well-known limitations of EMG signals, such as susceptibility to electrode displacement, perspiration, and intersubject variability, the IMU-based approach offers a more robust, noninvasive, and practical solution for real-world fatigue monitoring applications.

V. Conclusion

In this study, a real-time system that predicts bicep muscle fatigue during dumbbell curl exercises was developed. Real-time performance for models predicting muscle fatigue is crucial to our goal of preventing MSDs in occupational health. Since MSDs are most often the result of prolonged overexhaustion of muscles, such a model can be integrated into mobile devices (e.g., smartphones and watches) to give users live feedback about their level of muscle fatigue, including reminders on when to take breaks. Furthermore, with growing advances in occupational exoskeletons [45], [46], a model which operates in real-time can be integrated with these exoskeletons to allow for dynamic adjustment of assistive levels, i.e., employing adjustable assistive power depending on the users' level of muscle fatigue.

Moreover, the numerical range of 1–7 for fatigue predictions overcomes the practical limitations associated with discrete classification stages. For instance, one user may wish to enable exoskeleton assistance starting at level 4 fatigue, while another user may prefer to enable assistance starting at level 2 fatigue. If only a single category "low" is specified, the subjective fatigue perceptions of individuals are not taken into account. Therefore, future work is to reframe the modeling of muscle fatigue from a regression perspective, as demonstrated in this study.

Considering the results obtained in this study, the next step is to apply the framework proposed in this study to different physical activities targeting different muscle groups. For activities that do not exhibit a direct like-sinusoidal movement pattern, such as walking, other time series data segmentation methods will be needed.

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